CS 3244
Machine Learning

Week 8 Post-Class

Backpropagation, Universal Approximation of NN and Regularizing NNs

8 post

Kan Min-Yen

Ang Yi Zhe

Background Photo credits: Rafiq Mirza, Luan Ahn, Rosmarie Voegtli @ Flickr

Recap and Forecast

Understand how validation and regularization complement to address overfitting:

- **Regularization:** a means of restraining the model.
  - L^p norm forms of regularization.

$$J_{test}(h) = J_{train}(h) + \text{overfit penalty}$$

- **Validation**, as a reality check by peeking (at the bottom line).
  - Cross Validation variants: k-fold or leave one out.

Learning outcomes for this week

- Describe the basic idea of Artificial Neural Networks (ANN or NN):
  - Explain how neural networks combine linear units with a non-linear activation function;
  - Explain the topology of a NN;
- Name potential problems with NNs;
- Describe how a neural network is trained using the backpropagation algorithm;
- Name NN regularization techniques;
- Obtain practical experience constructing NNs using PyTorch.

Back Propagation

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Recap: Gradient Descent

GD minimizes:

$$J_{train}(\Theta) = \frac{1}{m}\sum_{j=1}^{m} j(h(\boldsymbol{x}^{(j)}), y^{(j)})$$

By iterative steps along $-\nabla J_{train}$:

$$\Delta\Theta = -\eta\nabla J_{train}(\Theta)$$

Batch GD: $-\nabla J_{train}$ is based on all examples $(\mathbf{X}, \boldsymbol{y})$

SGD: $-\nabla J_{train}$ is based on one example at a time $(\boldsymbol{x}^{(*)}, y^{(*)})$

$$\mathbb{E}_*\left[-\nabla j(h(\boldsymbol{x}^{(*)}), y^{(*)})\right] = -\nabla J_{train}$$

Applying SGD

All the weights $\Theta = \left\{\Theta_{i,j}^{[l]}\right\}$ determine $h(\boldsymbol{x})$

Error on example $\left(\boldsymbol{x}^{(*)}, y^{(*)}\right)$ is

$$j\left(h\left(\boldsymbol{x}^{(*)}\right), y^{(*)}\right) = j(\Theta)$$

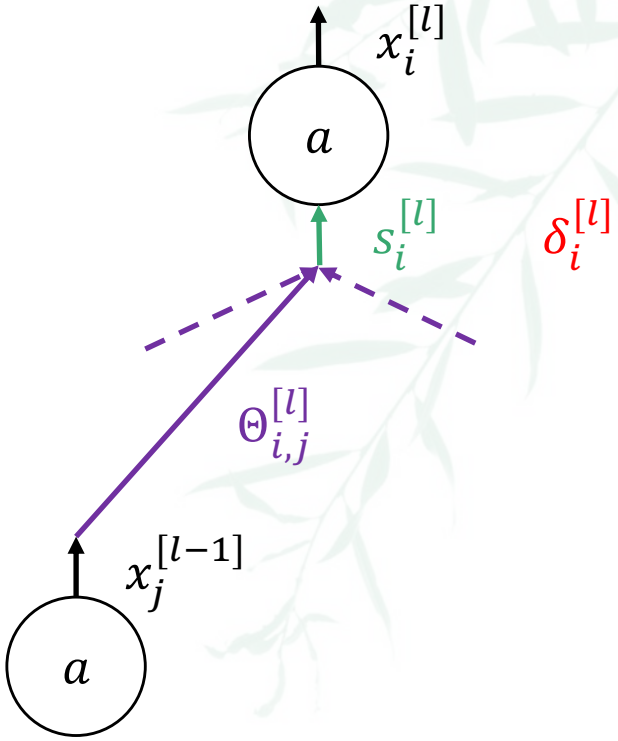
To implement SGD, we need the gradient:

$\nabla j(\Theta)$: $\dfrac{\partial j(\Theta)}{\partial\, \Theta_{i,j}^{[l]}}$ for all i, j, l

Computing $\frac{\partial\ j(\Theta)}{\partial\ \Theta_{i,j}^{[l]}}$

We can evaluate $\frac{\partial\ j(\Theta)}{\partial\ \Theta_{i,j}^{[l]}}$ one by one: analytically or numerically

A trick for efficient computation:

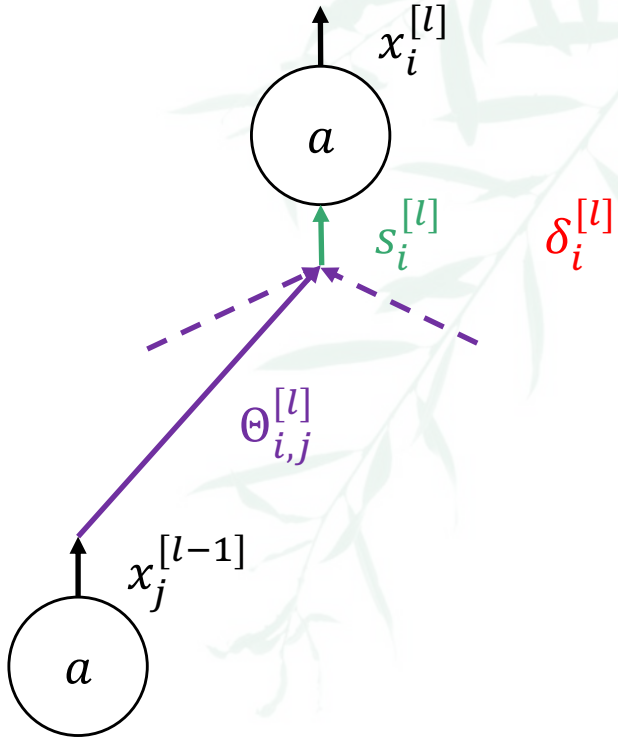


Computing $\frac{\partial\, j(\Theta)}{\partial\, \Theta_{i,j}^{[l]}}$

We can evaluate $\frac{\partial\, j(\Theta)}{\partial\, \Theta_{i,j}^{[l]}}$ one by one: analytically or numerically

A trick for efficient computation:

$$\frac{\partial\, j(\Theta)}{\partial\, \Theta_{i,j}^{[l]}} = \frac{\partial\, j(\Theta)}{\partial\, s_i^{[l]}} \times \frac{\partial\, s_i^{[l]}}{\partial\, \Theta_{i,j}^{[l]}}$$



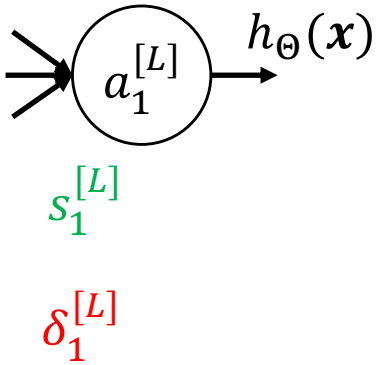
We have $\frac{\partial\, s_i^{[l]}}{\partial\, \Theta_{i,j}^{[l]}} = x_j^{[l-1]}$

We only need: $\frac{\partial\, j(\Theta)}{\partial\, s_i^{[l]}} = \delta_i^{[l]}$

δ for the final layer

$$\delta_i^{[l]} = \frac{\partial j(\Theta)}{\partial s_i^{[l]}}$$

For the final layer $l = L$ and $i = 1$:



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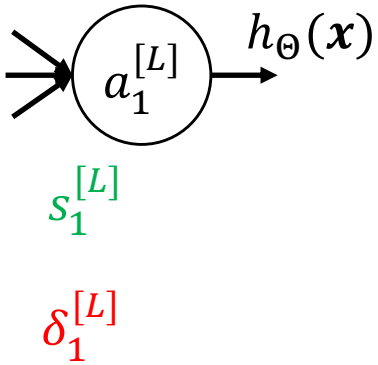
For the final layer $l = L$ and $i = 1$:

$$\delta_1^{[L]} = \frac{\partial j(\Theta)}{\partial s_1^{[L]}}$$



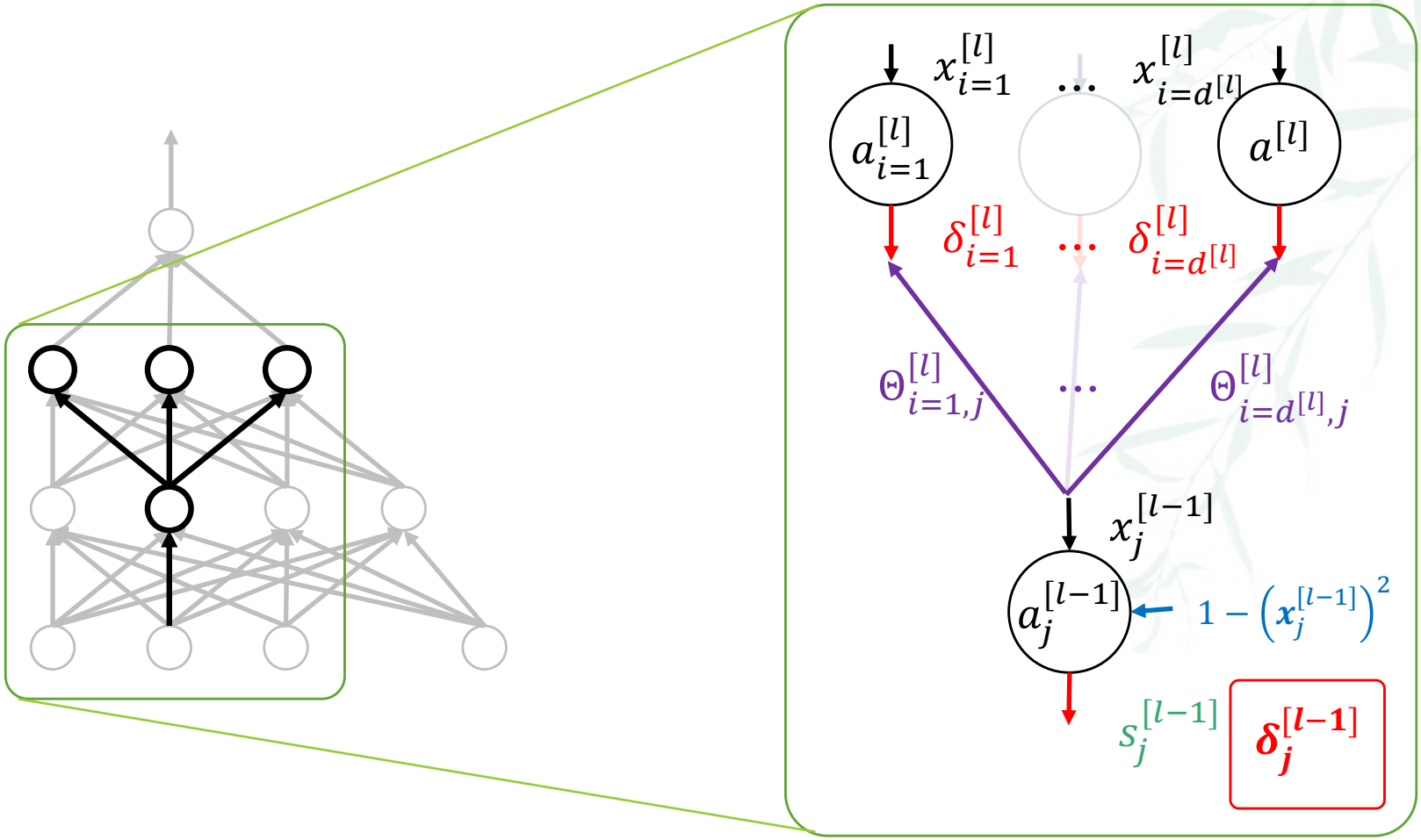
$$j(\Theta) = j(h(\boldsymbol{x}^{(*)}), y^{(*)}) = \left(a_1^{[L]} - y^{(*)}\right)^2$$

$$a_1^{[L]} = g\left(s_1^{[L]}\right)$$

$$g'(s) = 1 - g^2(s) \text{ for tanh}$$

Back Propagation of δ

$$\boxed{\boldsymbol{\delta_j^{[l-1]}}} = \frac{\partial\, j(\Theta)}{\partial\, s_j^{[l-1]}}$$



Back Propagation of δ

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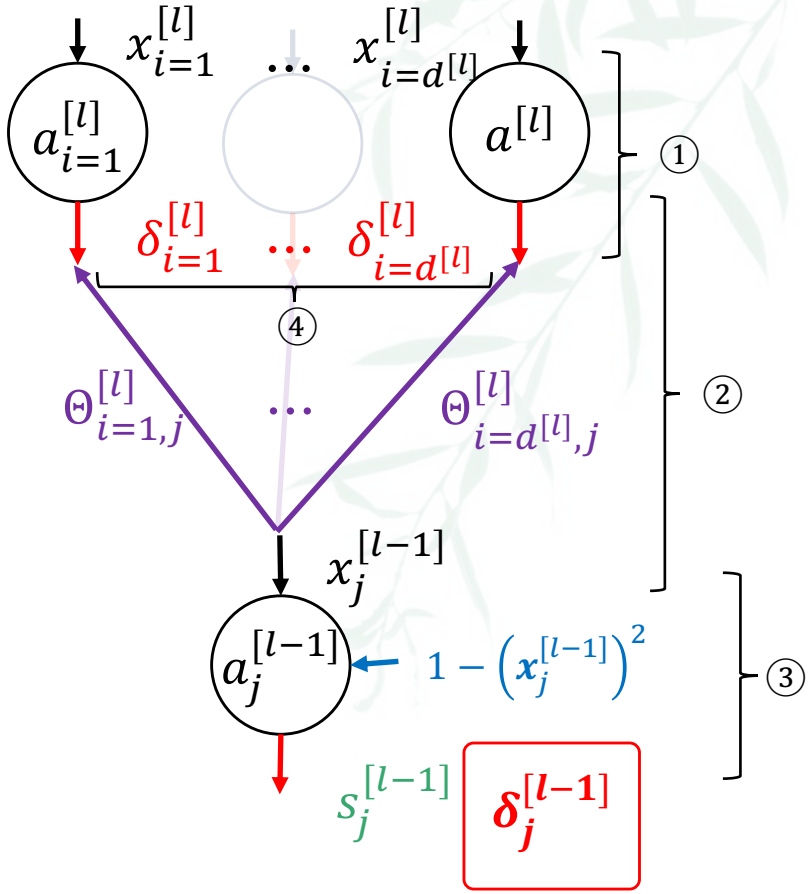


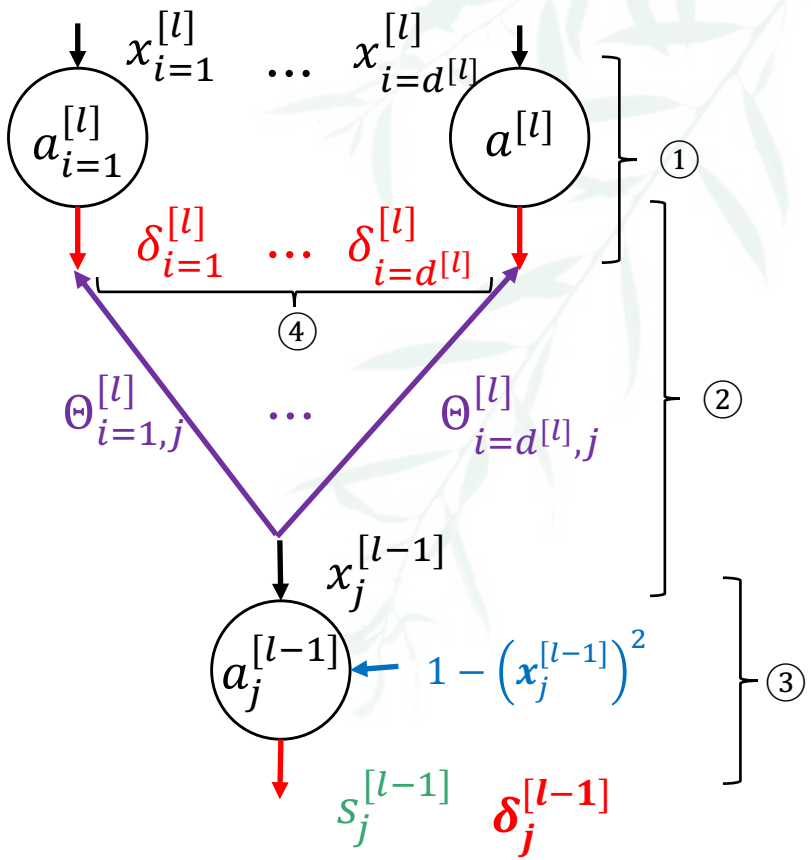
Back Propagation of δ

The view from a single unit $\boldsymbol{\delta_j^{[l-1]}} = \dfrac{\partial\, j(\Theta)}{\partial\, s_j^{[l-1]}}$

$$= \underbrace{\sum_i^{d^{[l]}}}_{④} \underbrace{\frac{\partial\, j(\Theta)}{\partial\, s_i^{[l]}}}_{①} \times \underbrace{\frac{\partial\, s_i^{[l]}}{\partial\, x_j^{[l-1]}}}_{②} \times \underbrace{\frac{\partial\, x_j^{[l-1]}}{\partial\, s_j^{[l-1]}}}_{③}$$

$$= \sum_i^{d^{[l]}} \delta_i^{[l]} \times \Theta_{i,j}^{[l]} \times \underbrace{g'(s_j^{[l-1]})}_{⑤}$$

$$\boldsymbol{\delta_j^{[l-1]}} = \underbrace{\left(1 - \left(x_j^{[l-1]}\right)^2\right)}_{⑤} \sum_{i=1}^{d^{[l]}} \Theta_{i,j}^{[l]} \delta_i^{[l]}$$



Backpropagation Algorithm

initialize all weights $\Theta_{i,j}^{[l]}$ ~~as $\vec{0}$~~ at random

for $t = 0, 1, 2, \ldots$ do

pick $(\boldsymbol{x}^{(*)}, y^{(*)}) \in \{1,2,\ldots,m\}$

Forward: Compute all $x^{[l]}$

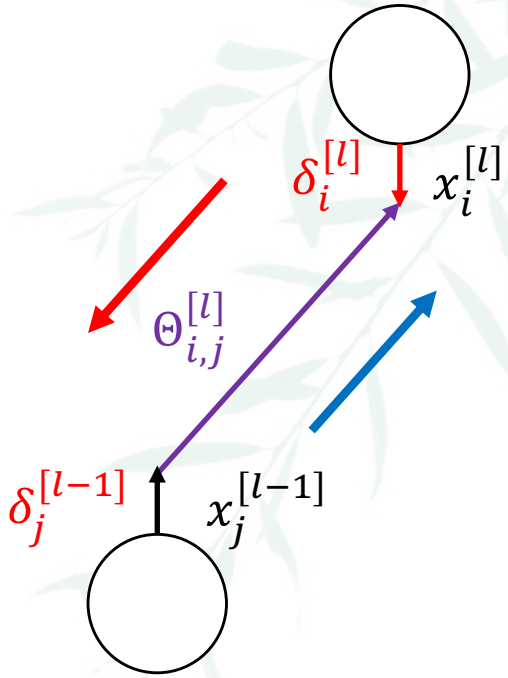
Backward: Compute all $\delta^{[l]}$

Update the weights:

$$\Theta_{i,j}^{[l]} \leftarrow \Theta_{i,j}^{[l]} - \eta\, x_j^{[l-1]} \delta_i^{[l]}$$

Iterate to the next step until termination

Return the final weights $\Theta_{i,j}^{[l]}$



Stuck? Try walking through the online tutorial by Matt Mazur.

Learning Nonlinear Transforms

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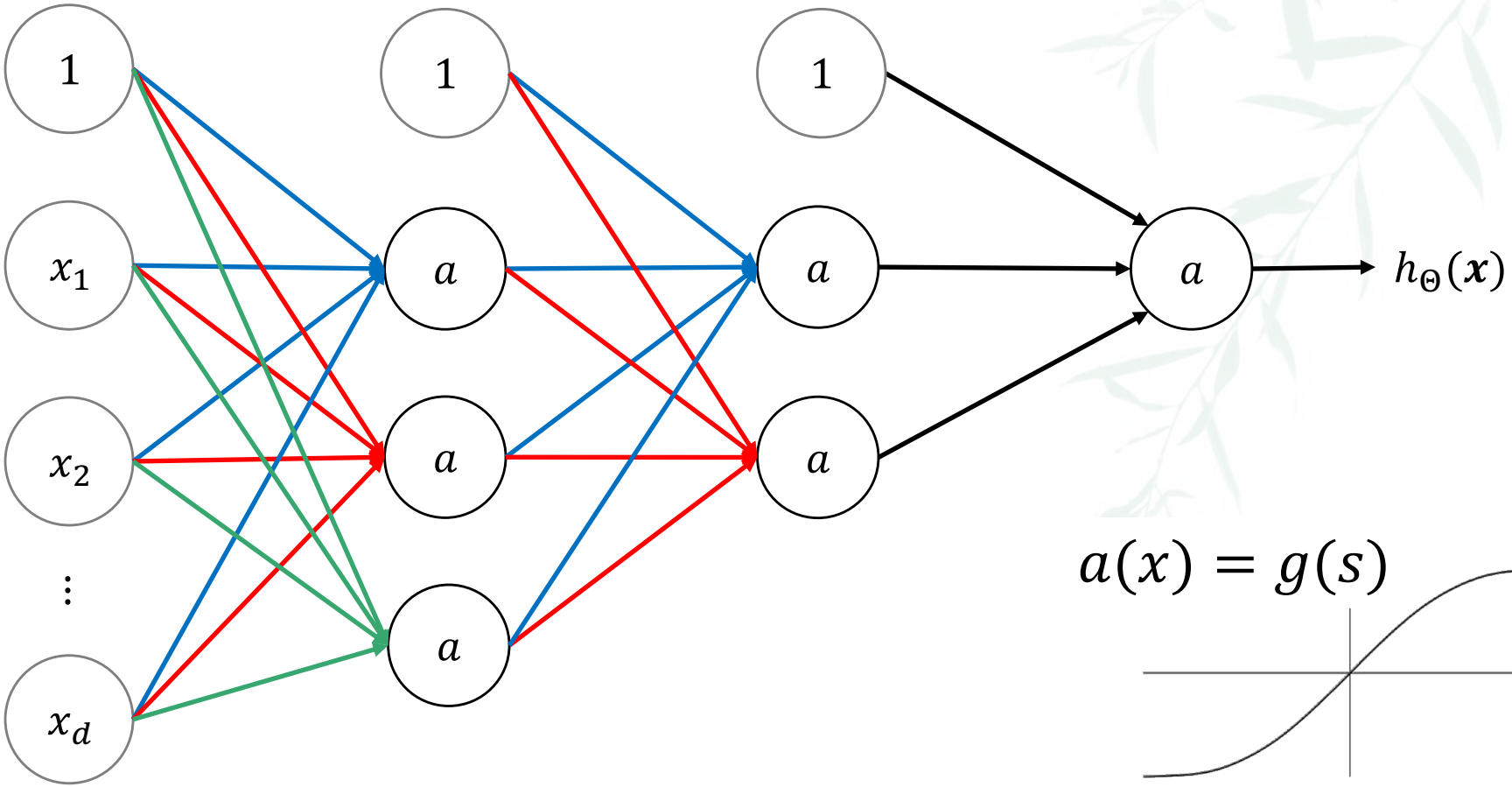
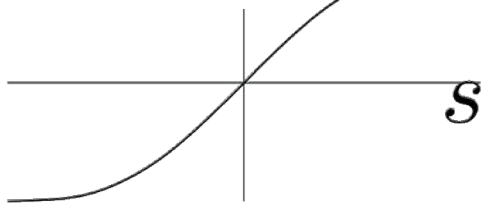
Hidden Layers

learned nonlinear transform

Interpretation?



$a(x) = g(s)$



FeedForward Example

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Forward Propagation

Without recurrence: Easy Peasy

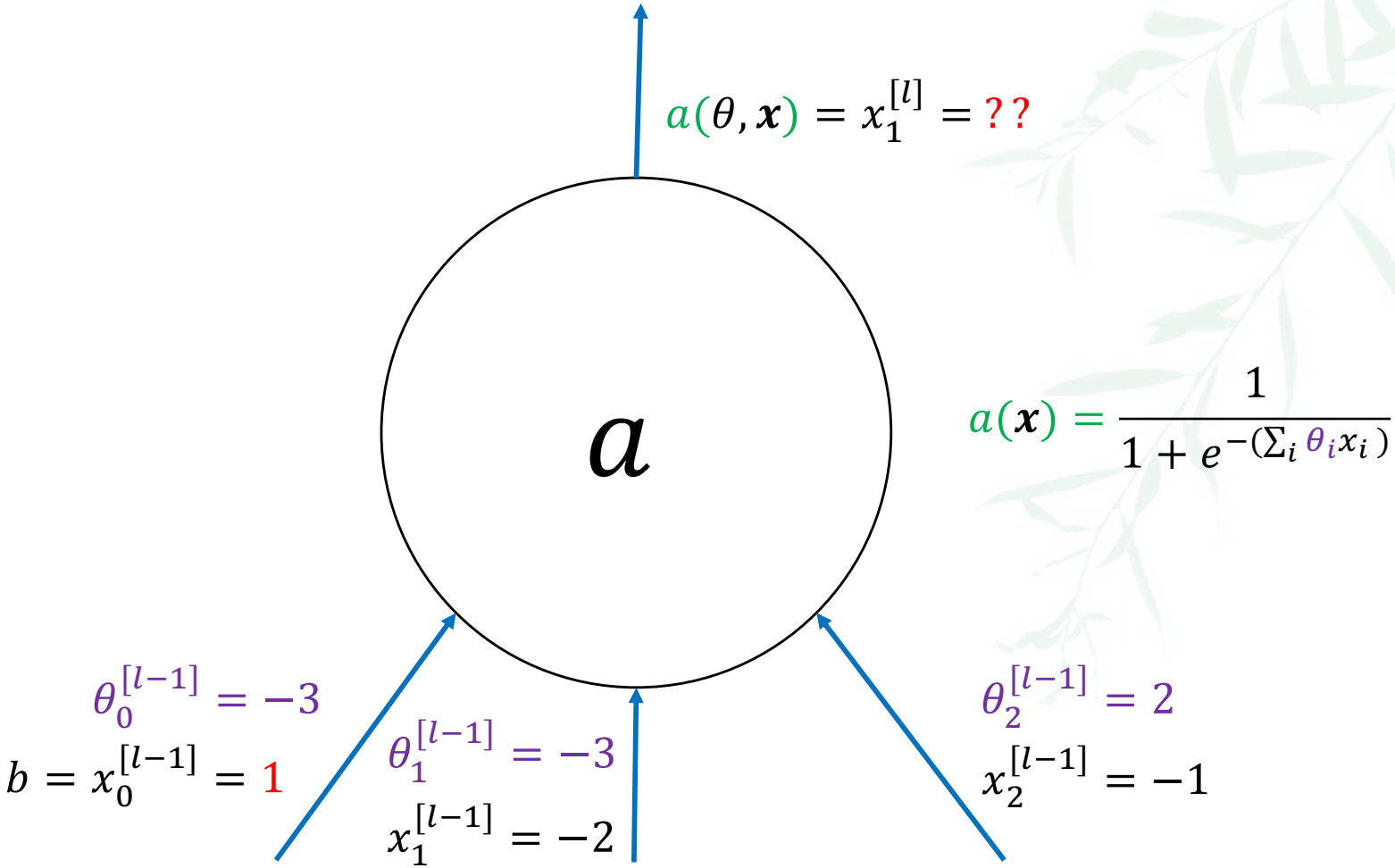
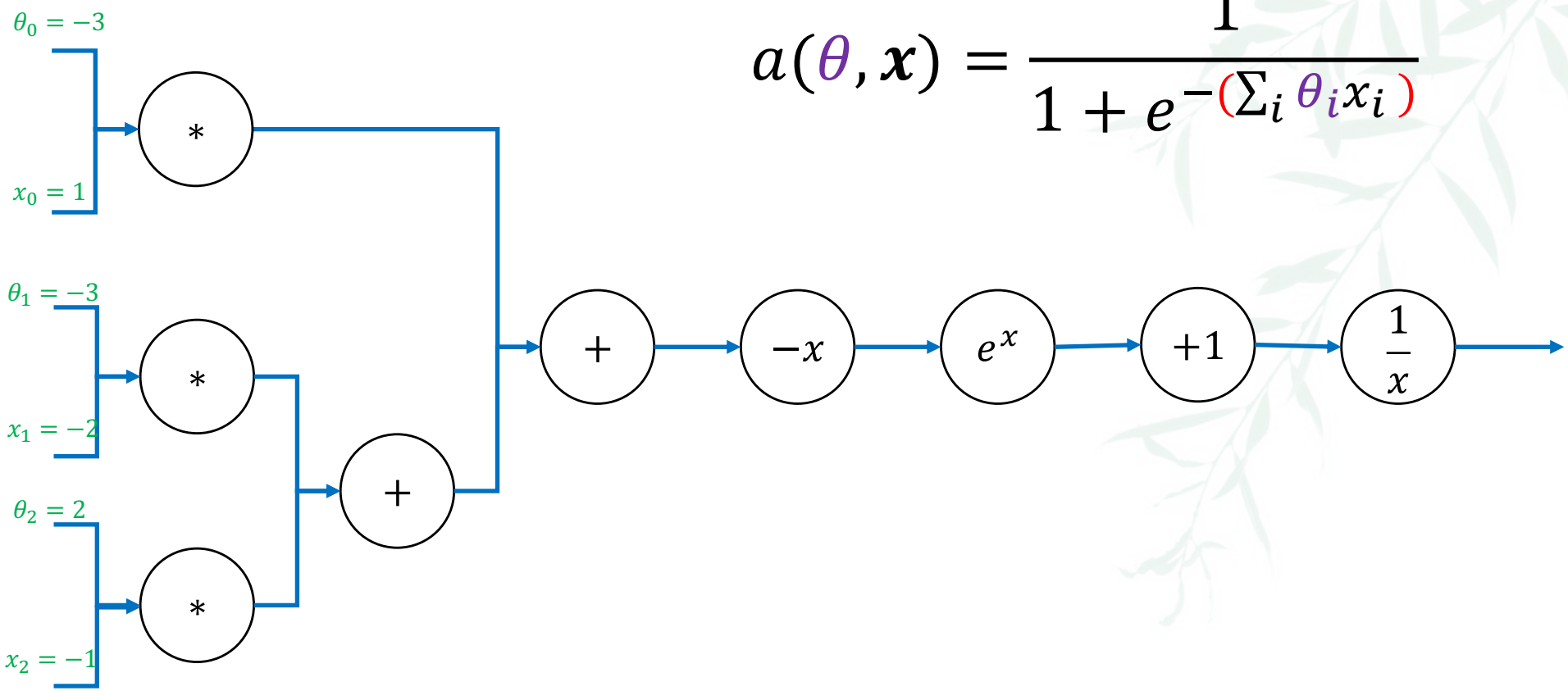


Diagram credits for this section: Fei-Fei Li, Justin Johnson and Serena Yeung (Stanford CS231n)

Feedforward Sigmoid Example

Envision calculations as part of a computation graph, with atomic operations.

$$a(\theta, \boldsymbol{x}) = \frac{1}{1 + e^{-(\sum_i \theta_i x_i)}}$$



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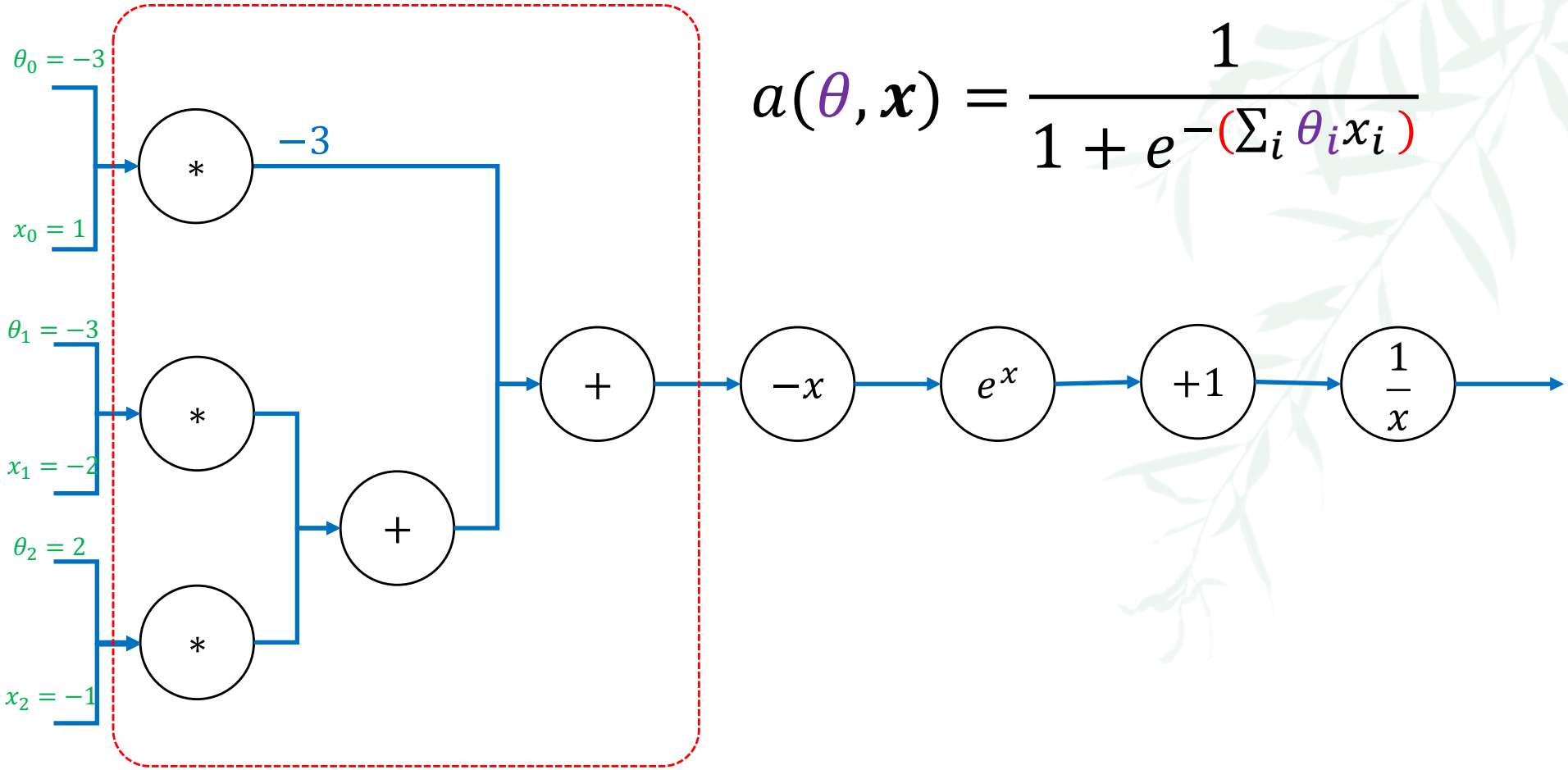
Feedforward Sigmoid Example



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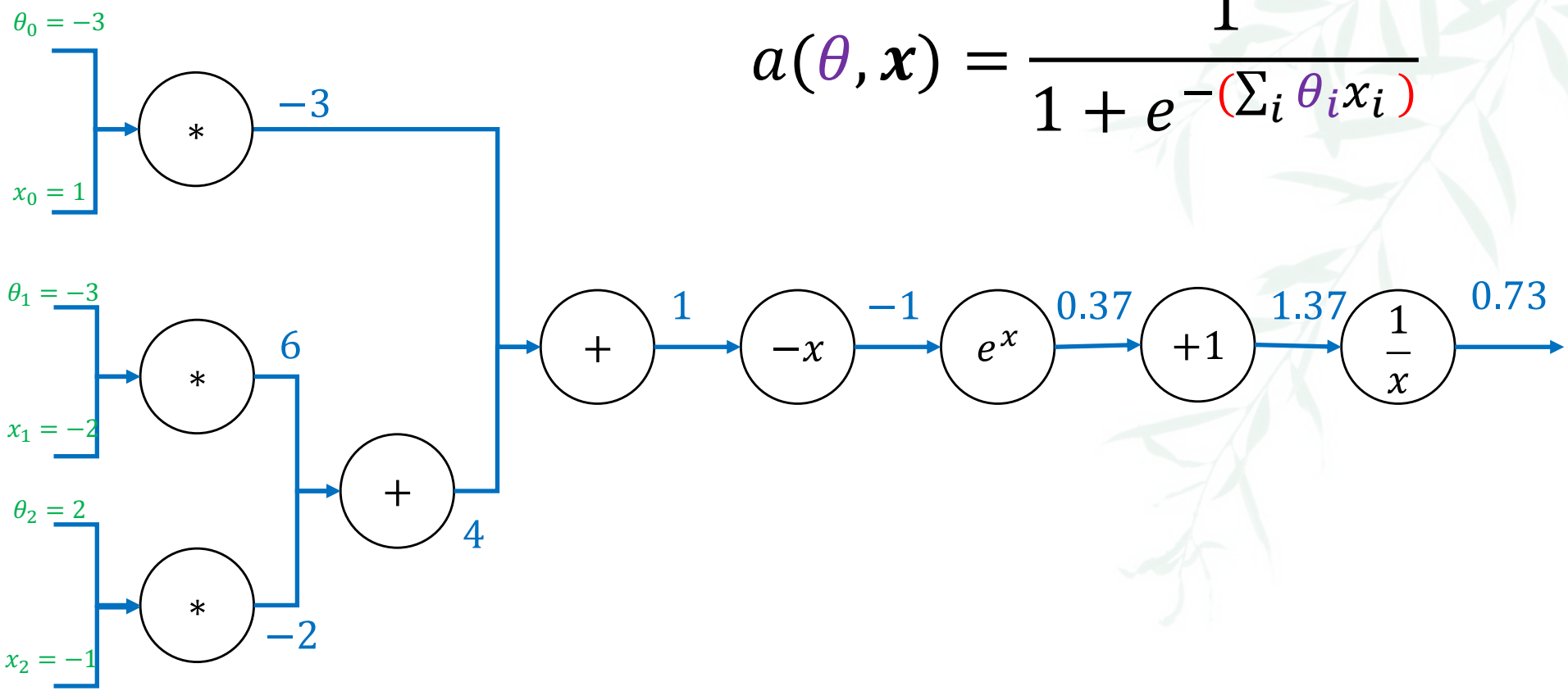

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Backprop Example

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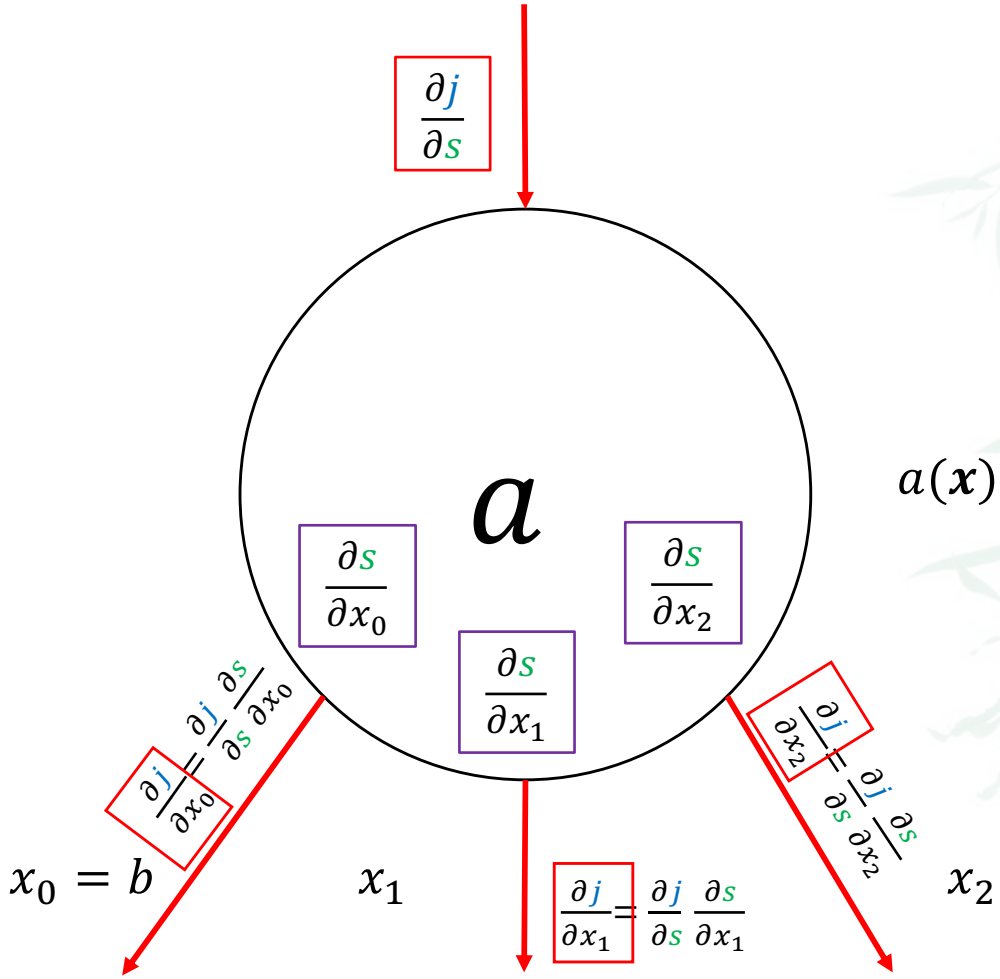
Chain Rule ≡ Backprop

Local Gradient: How the node views inputs with respect to outputs



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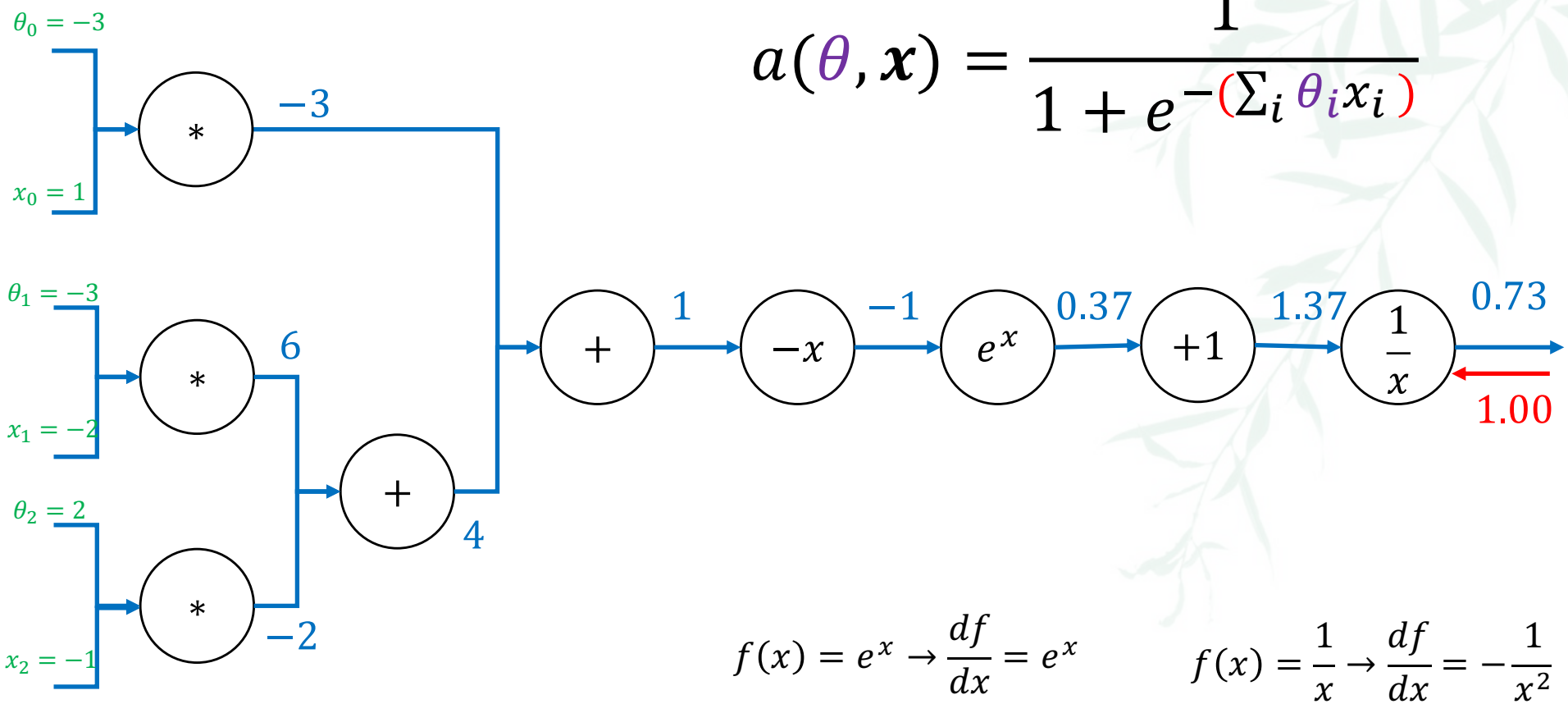

$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x \qquad f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -\frac{1}{x^2}$$

$$f = ab \rightarrow \frac{\partial f}{\partial a} = b, \frac{\partial f}{\partial b} = a \qquad f_a(x) = ax \rightarrow \frac{df}{dx} = a \qquad f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$$

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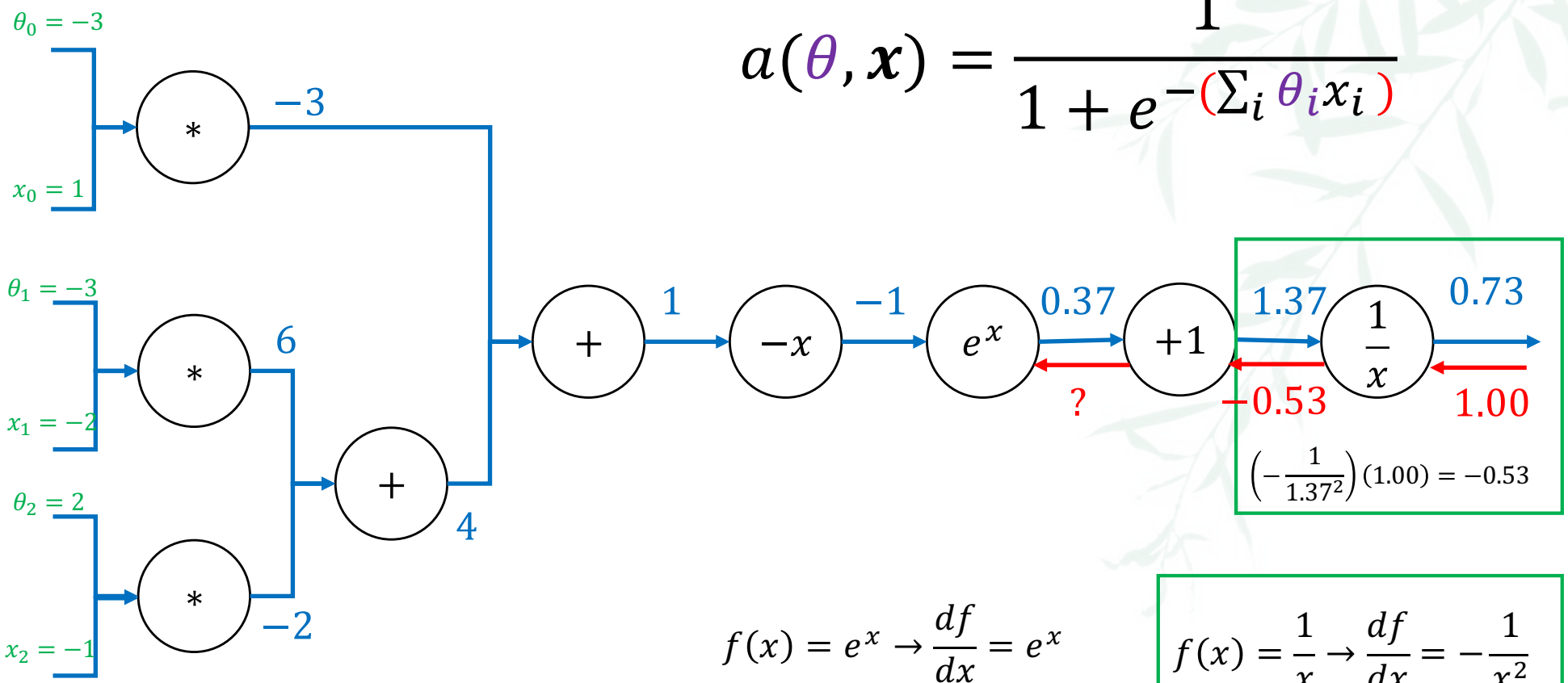

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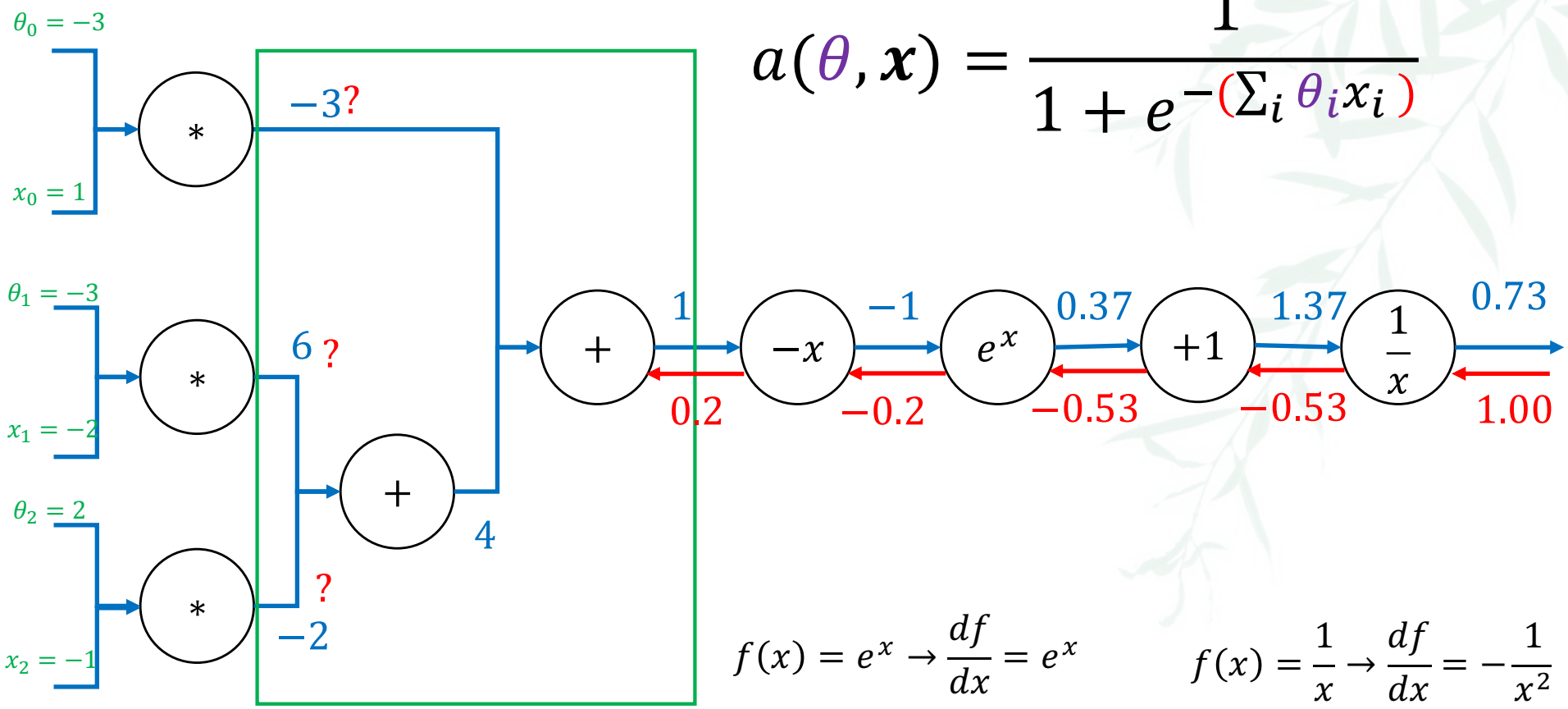

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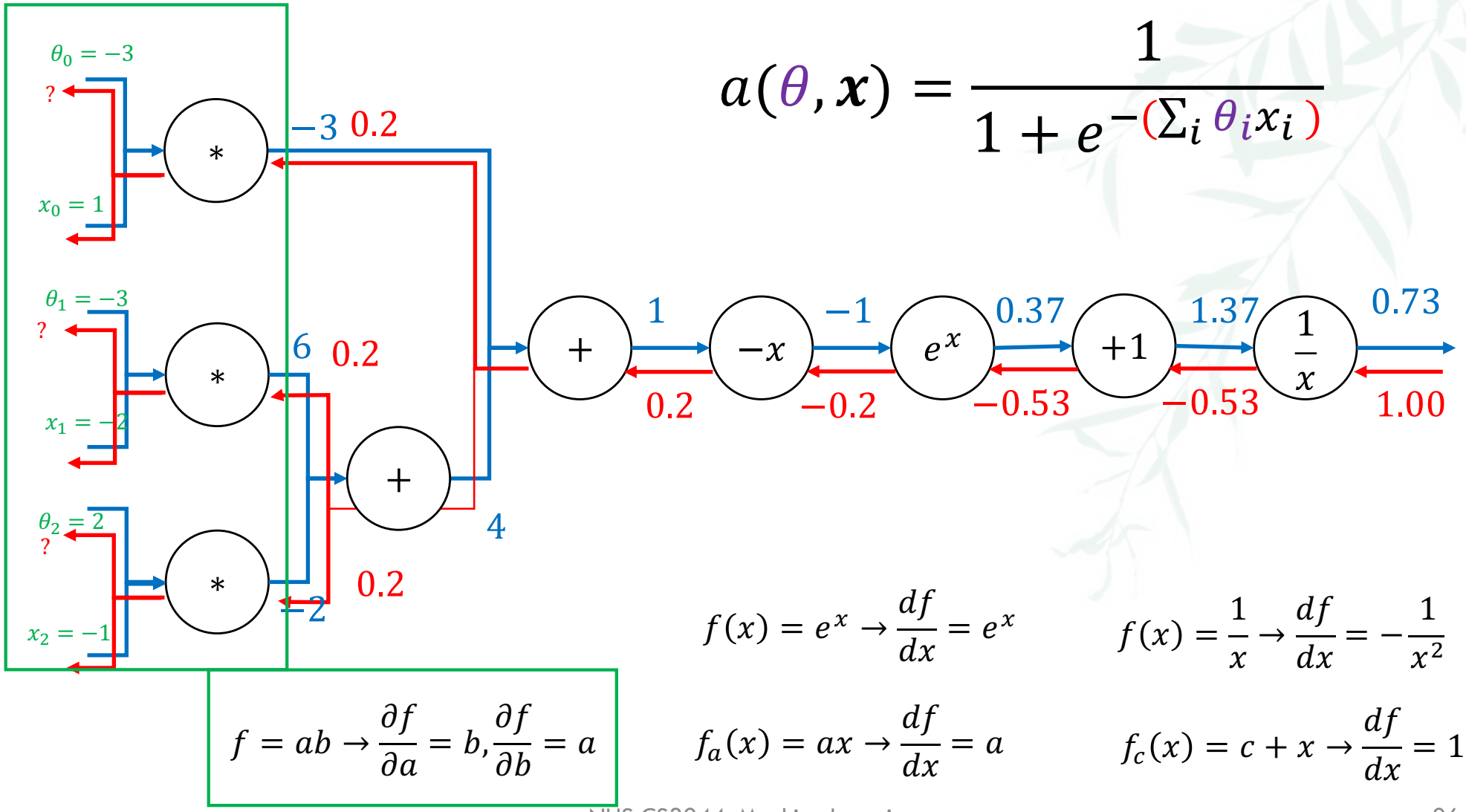

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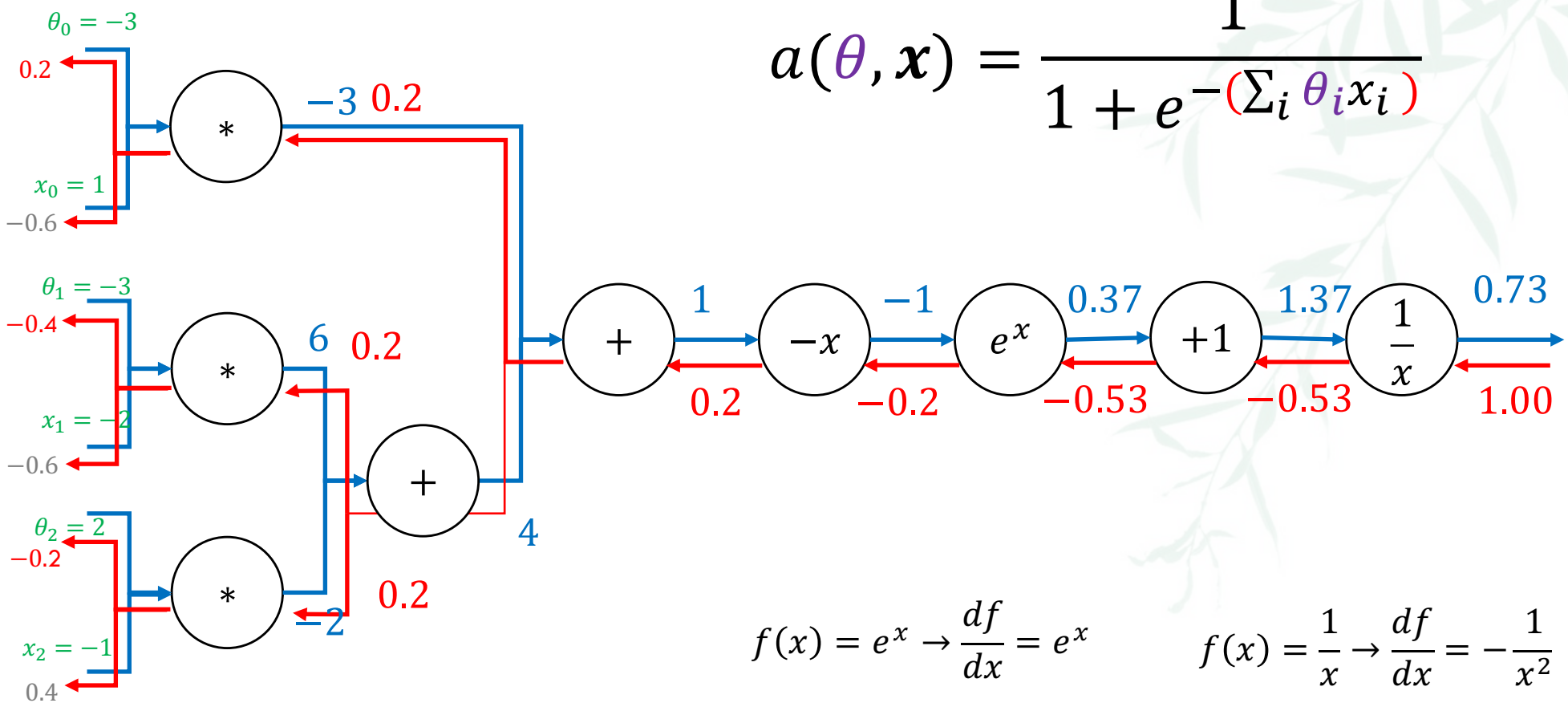

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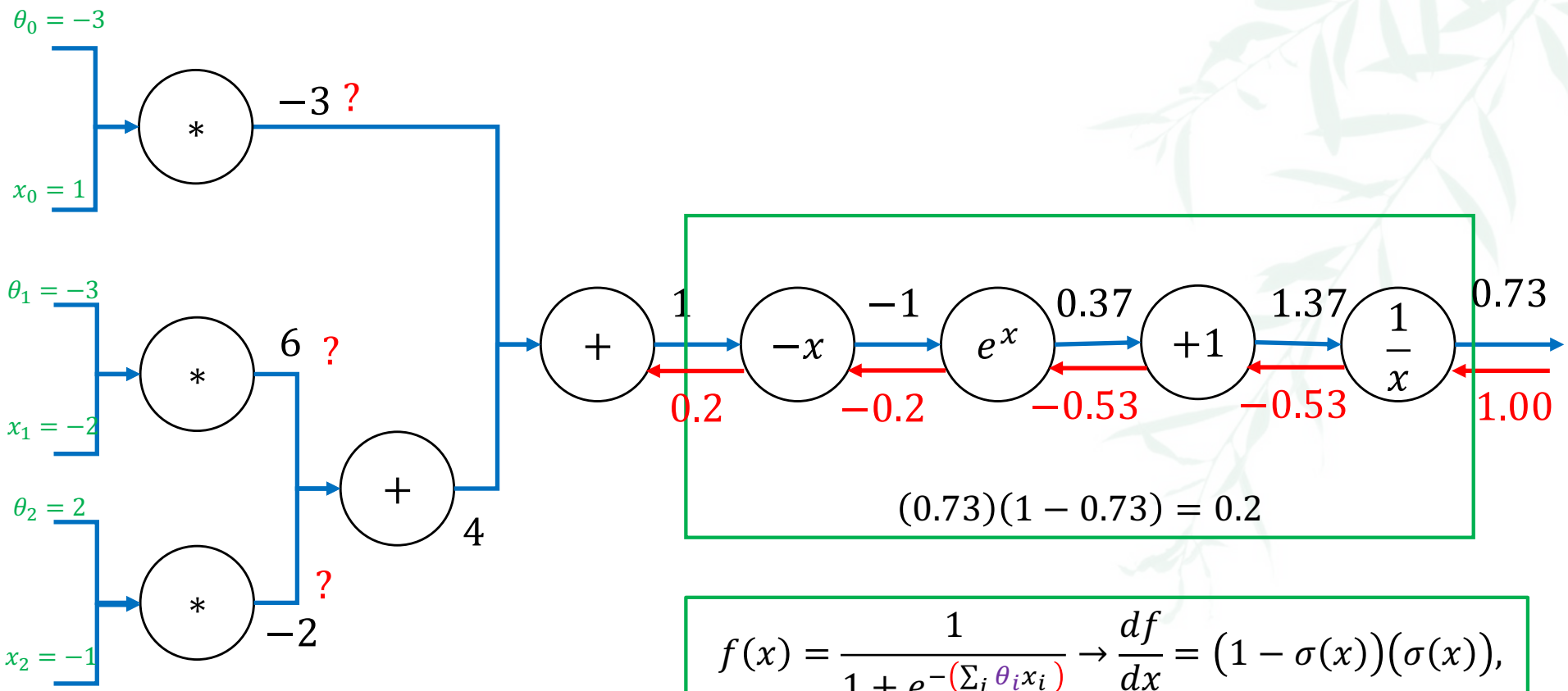
Sigmoid as a composition



$$f(x) = \frac{1}{1+e^{-(\Sigma_i \theta_i x_i)}} \rightarrow \frac{df}{dx} = \big(1-\sigma(x)\big)\big(\sigma(x)\big),$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$

Diagram credits for this section: Fei-Fei Li, Justin Johnson and Serena Yeung (Stanford CS231n)

Universal Approximation Theorem

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Summary: Universal Approximation

A single hidden layer NN of finite size can approximate any function

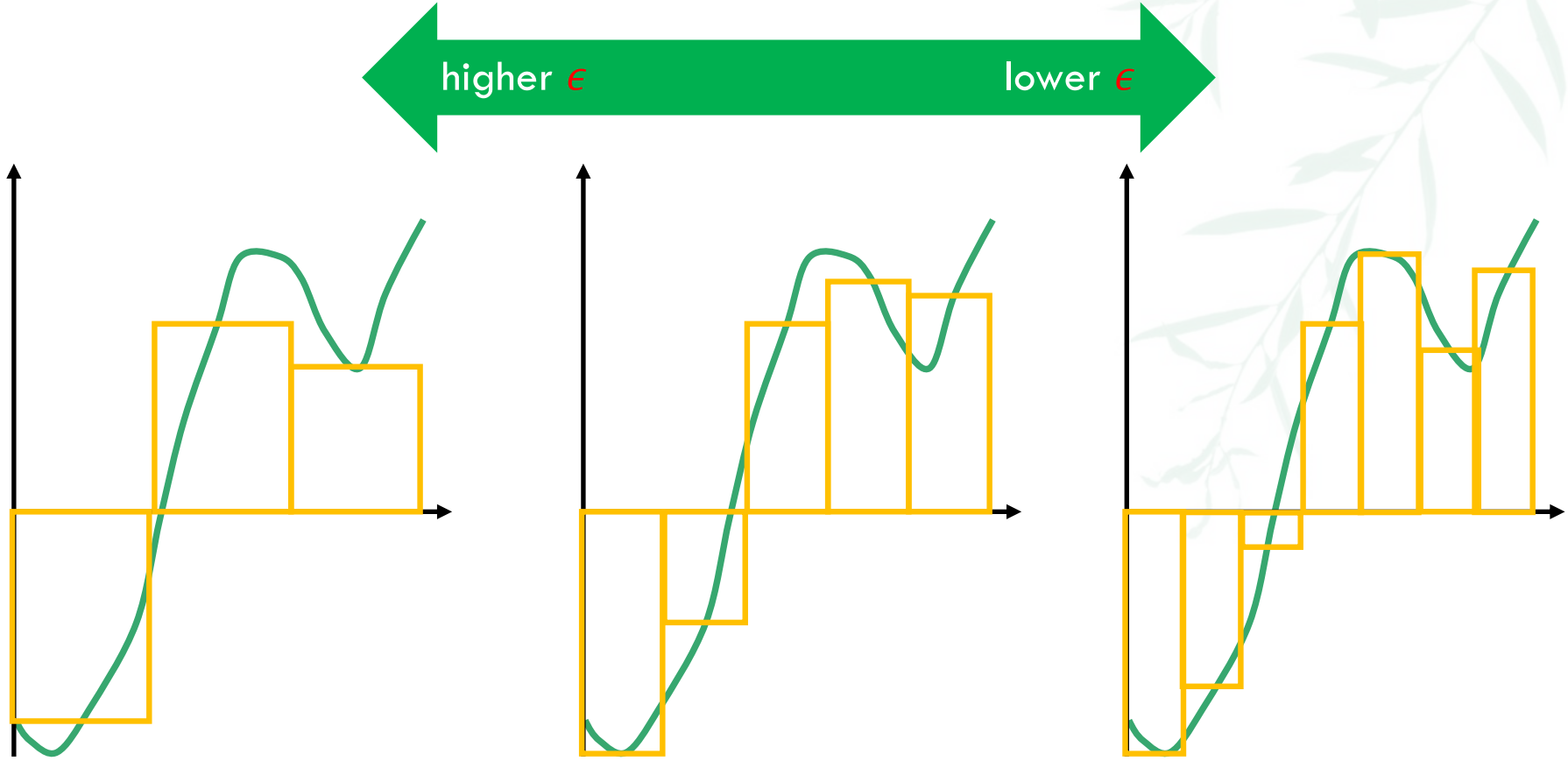
We can calculate $|h_\Theta(\boldsymbol{x}) - f(\boldsymbol{x})| < \epsilon$, for any ϵ that we wish. For lower ϵ, we'll need accordingly more units.

Even though we know that an NN exists that can compute $f(\boldsymbol{x})$, we may not have the means for its constructing or recognition.

Holds for the topology, non-linear activation function doesn't really matter.

A Graphical Proof

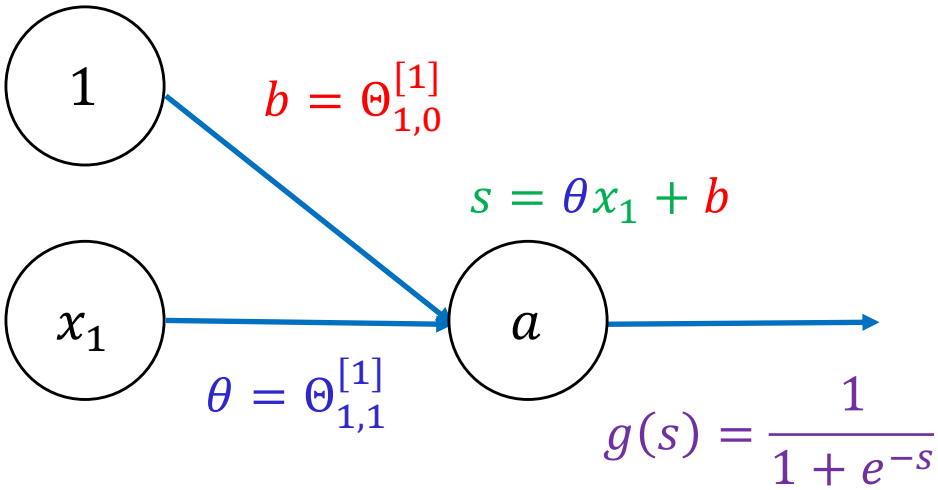
1. Build a bump function from linear units
2. Compose an arbitrary number of bumps to approximate $f(\boldsymbol{x})$.

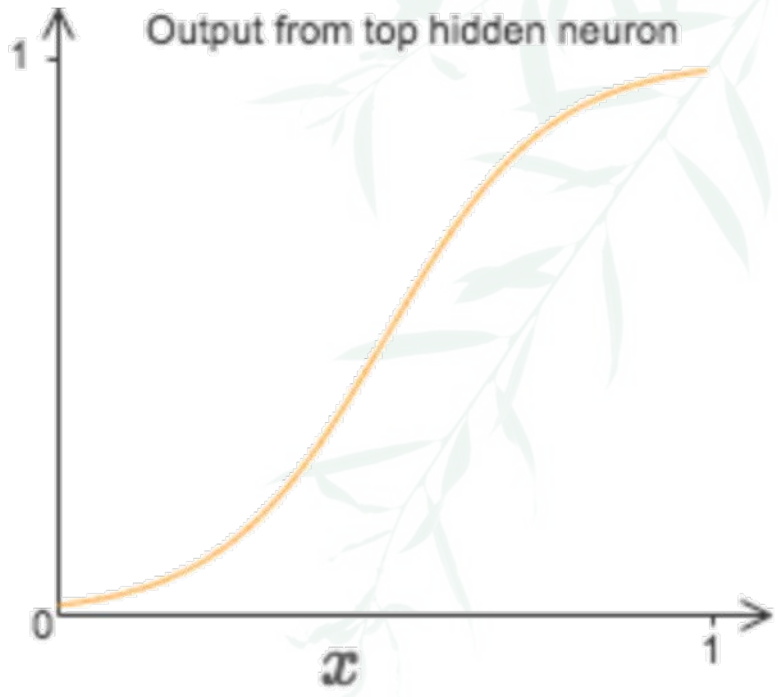


Adapted based on Michael A. Nielsen "NN and DL" 2015, Determination Press, CC By-NC 3.0

Recap: the single univariate unit

A single logistic unit computes a smooth $\mathcal{S}$-curve of the input.



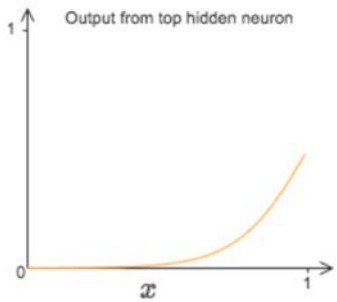


$g(x|\theta = 8, b = -4)$

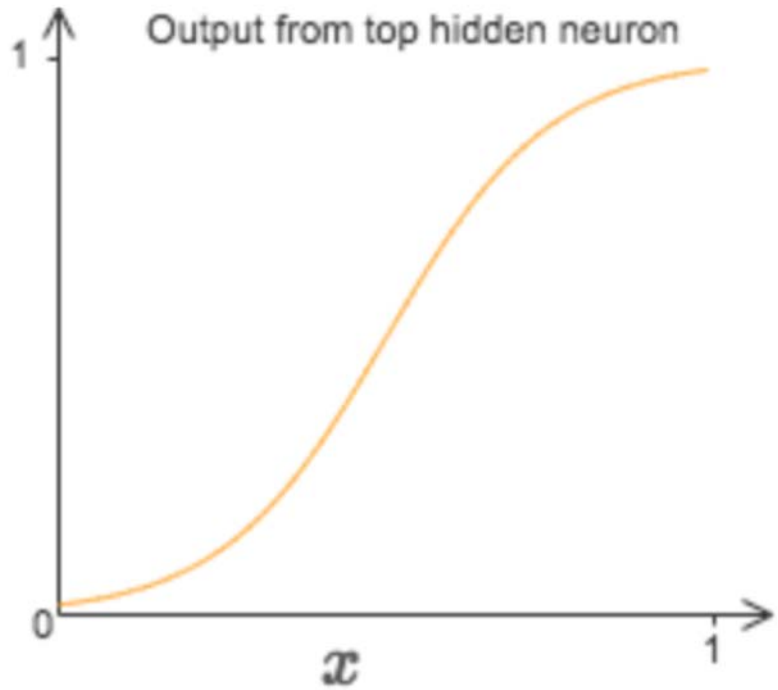
Adapted based on Michael A. Nielsen "NN and DL" 2015, Determination Press, CC By-NC 3.0

Varying θ and b

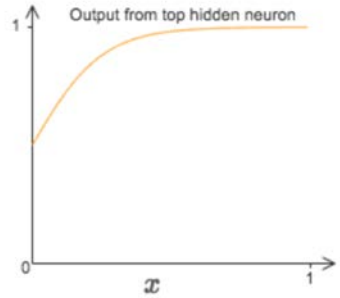
b varies the **shift** of the function



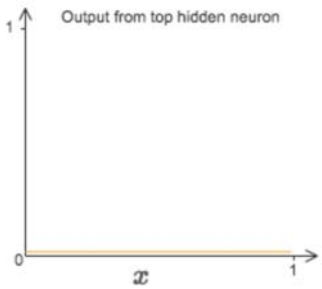
$g(x|\theta = 8, b = -8)$



$g(x|\theta = 8, b = -4)$



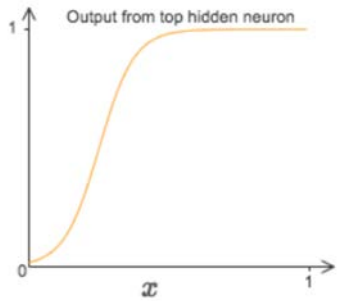
$g(x|\theta = 8, b = 0)$



$g(x|\theta = 0, b = -4)$

"linear"

θ varies the **slope** of the function



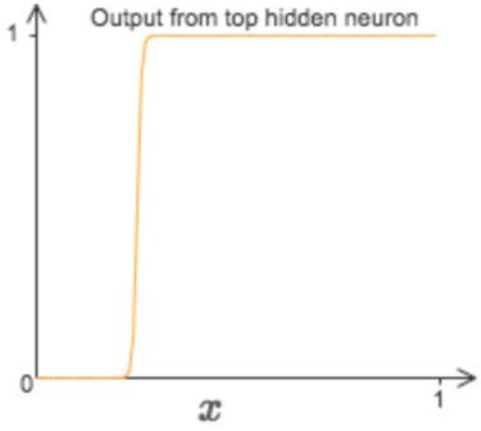
$g(x|\theta = 16, b = -4)$

"step"

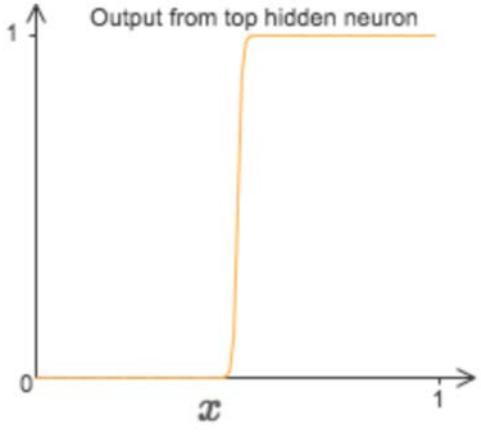
Adapted based on Michael A. Nielsen "NN and DL" 2015, Determination Press, CC By-NC 3.0

Let's work with step functions

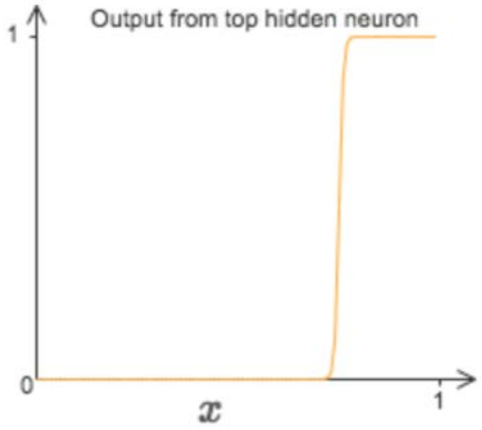
- Set θ high to get step-like output.
- Step occurs at $s = -\frac{b}{\theta}$.



$g(x|\theta = 200, b = -50)$



$g(x|\theta = 200, b = -100)$



$g(x|\theta = 200, b = -150)$

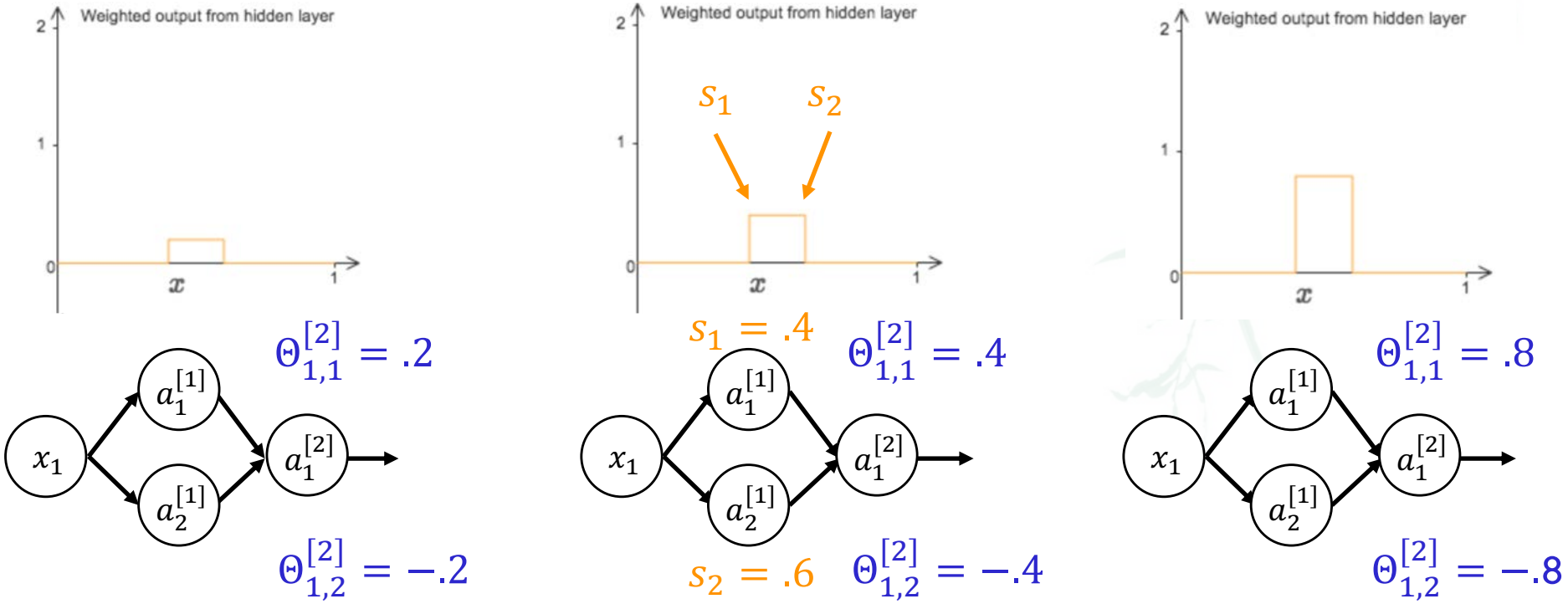
Adapted based on Michael A. Nielsen "NN and DL" 2015, Determination Press, CC By-NC 3.0

2 units: Computing a bump

One unit for step up and one unit for step down.

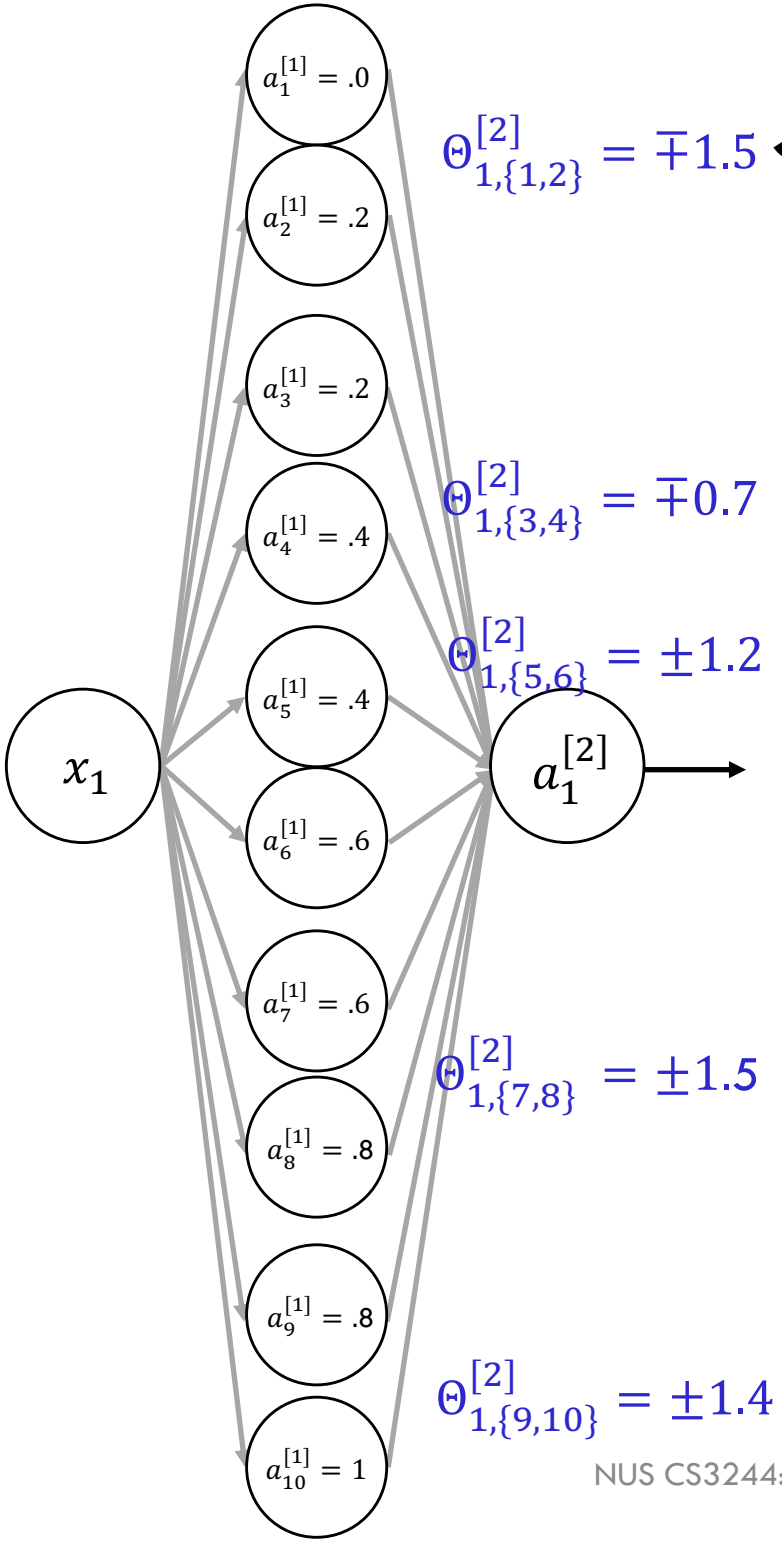
Weights at 2nd hidden layer additively compose both steps to form a bump.

Width determined by s_1 and s_2.



Adapted based on Michael A. Nielsen "NN and DL" 2015, Determination Press, CC By-NC 3.0

5 pairs of units



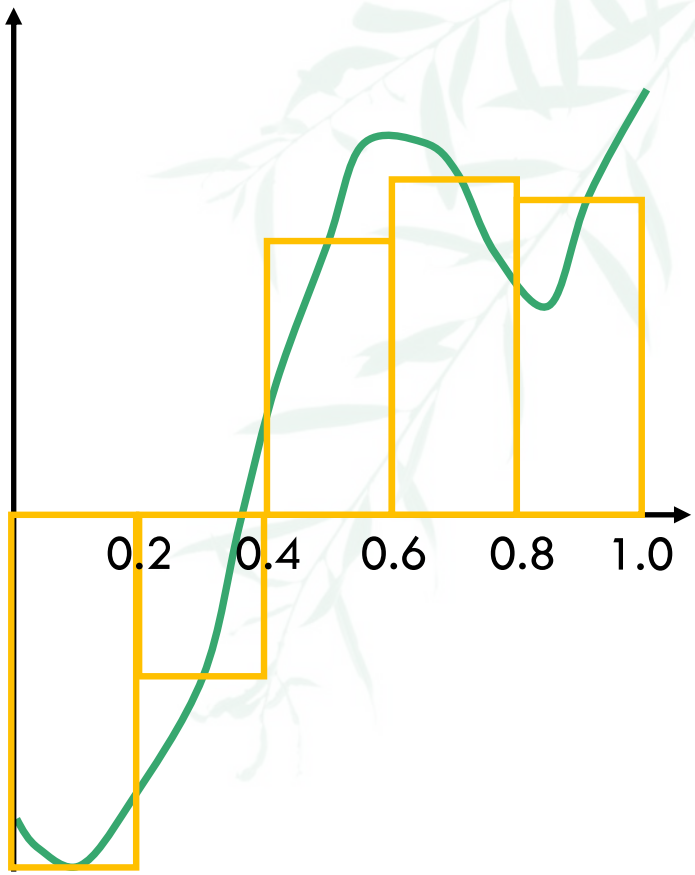
$\Theta^{[2]}_{1,\{1,2\}} = \mp 1.5$

$\Theta^{[2]}_{1,\{3,4\}} = \mp 0.7$

$\Theta^{[2]}_{1,\{5,6\}} = \pm 1.2$

$\Theta^{[2]}_{1,\{7,8\}} = \pm 1.5$

$\Theta^{[2]}_{1,\{9,10\}} = \pm 1.4$



Adapted based on Michael A. Nielsen "NN and DL" 2015, Determination Press, CC By-NC 3.0

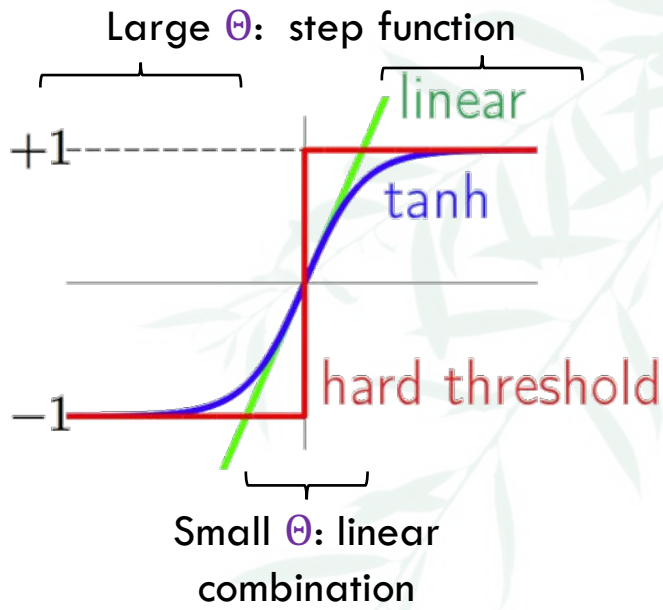
Regularizing NNs

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Regularization

1. Weight Decay: From linear to logical



2. Weight Elimination:

   Fewer weights ⟶ lower complexity, better generalization

Soft weight elimination:

$$\Omega(\Theta) = \sum_{i,j,l} \frac{\left(\Theta_{i,j}^{[l]}\right)^2}{1 + \left(\Theta_{i,j}^{[l]}\right)^2}$$

Small Θ pushed to zero; Big Θ left alone

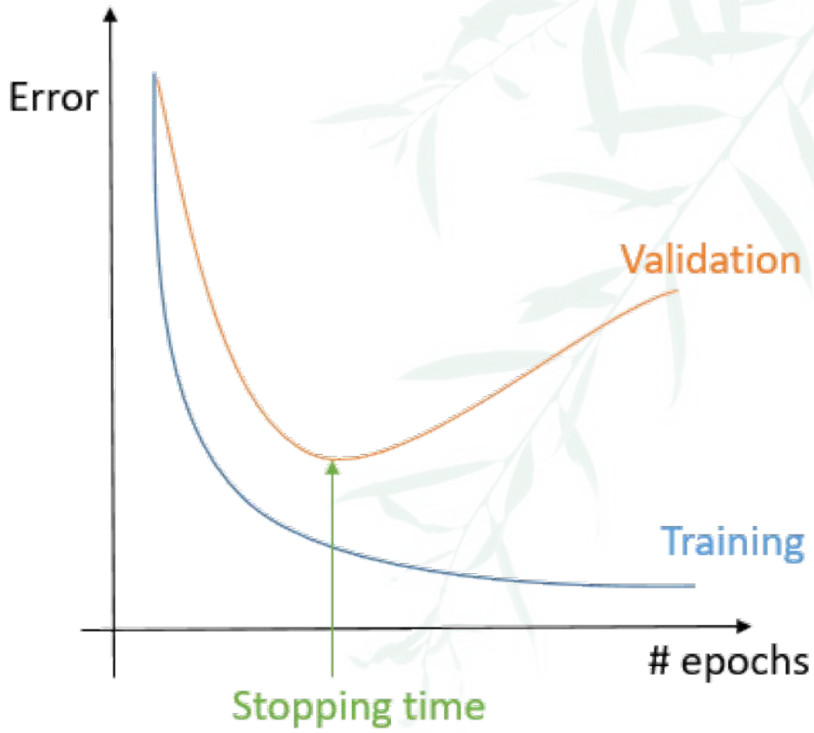
Early stopping as a regularizer

Regularization through the optimizer

Exploring the weight space per iteration/epoch

Budget $C \uparrow$ per iteration

When to stop? Validation!



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